Introduction to logistic regression

Agenda

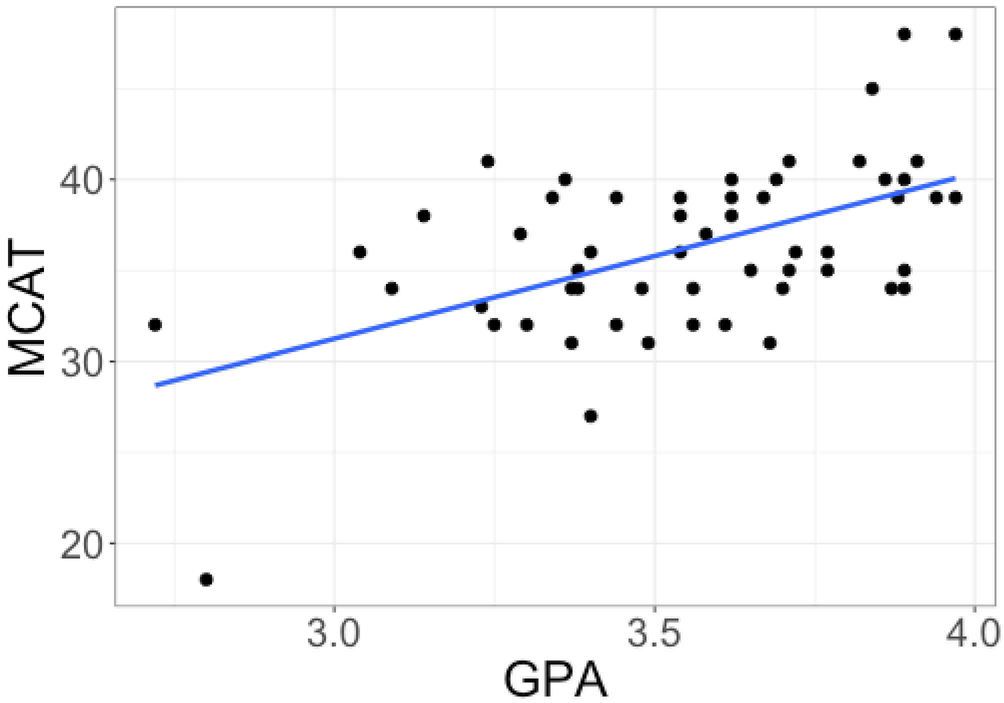
- Project 1, Part 4 due last day of class
  - There will be some time to work on the project during the last week of class
  - I have removed the oral presentation requirement and redistributed points among the project components
- Today: begin logistic regression

Data

55 students applying to medical school, with the following information on each student:

- `Acceptance`: whether the student was accepted to medical school (1 = accepted, 0 = denied)
- `GPA`: the student's college GPA
- `MCAT`: the student's MCAT score

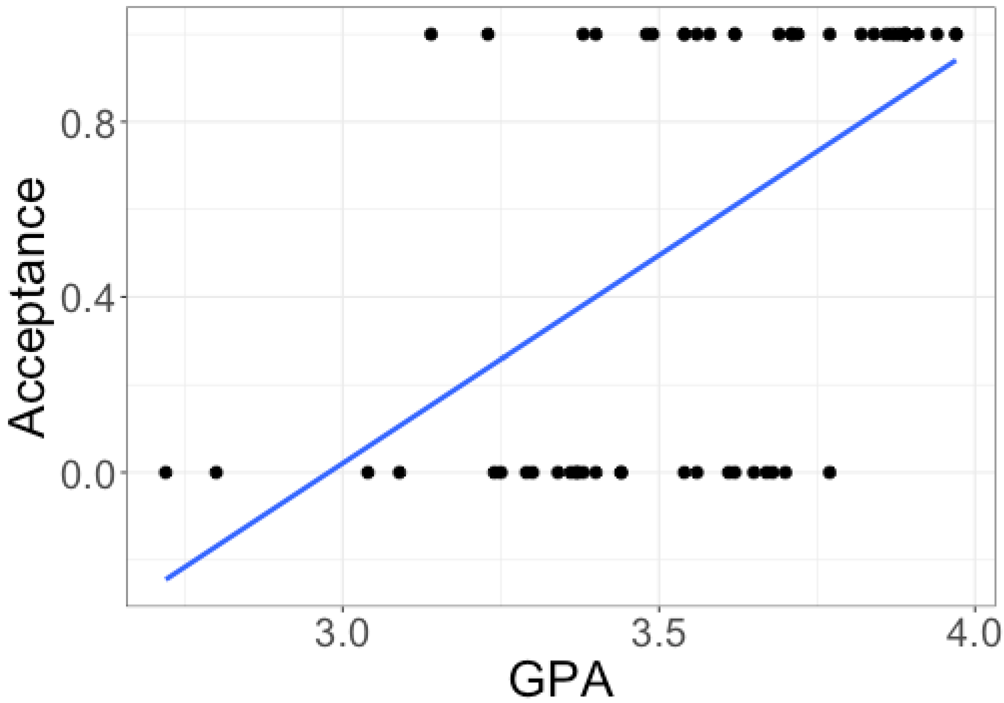
Linear regression



$$MCAT = \beta_0 + \beta_1 GPA + \varepsilon$$

> What if our response isn't quantitative?

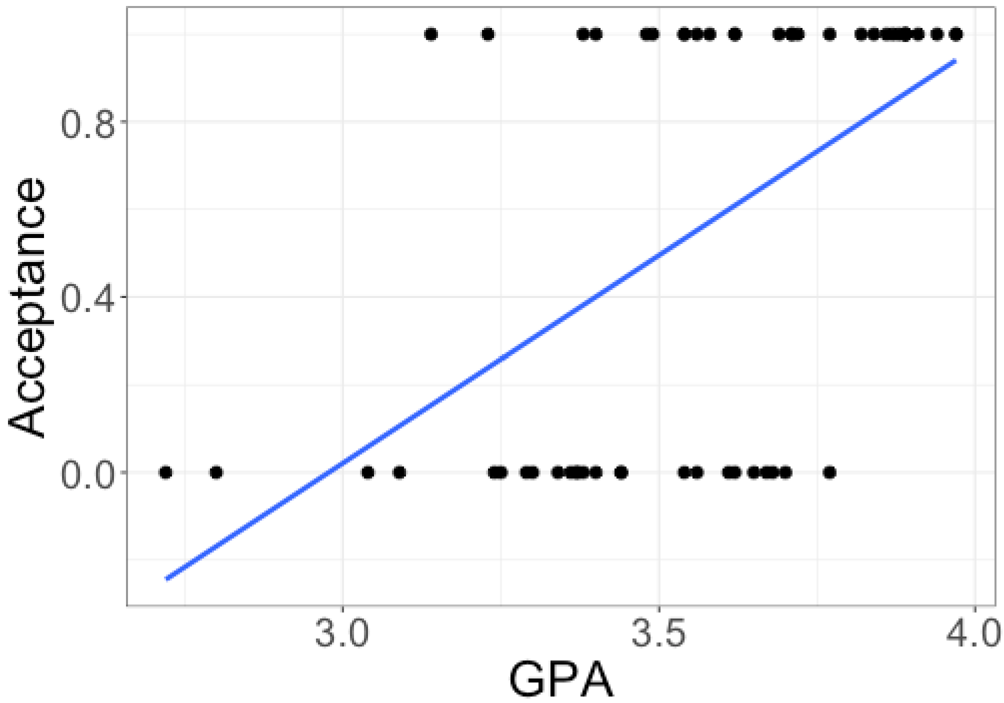
Binary response



$$Acceptance \stackrel{?}{=} \beta_0 + \beta_1 GPA + \varepsilon$$

> What are the issues with this model?

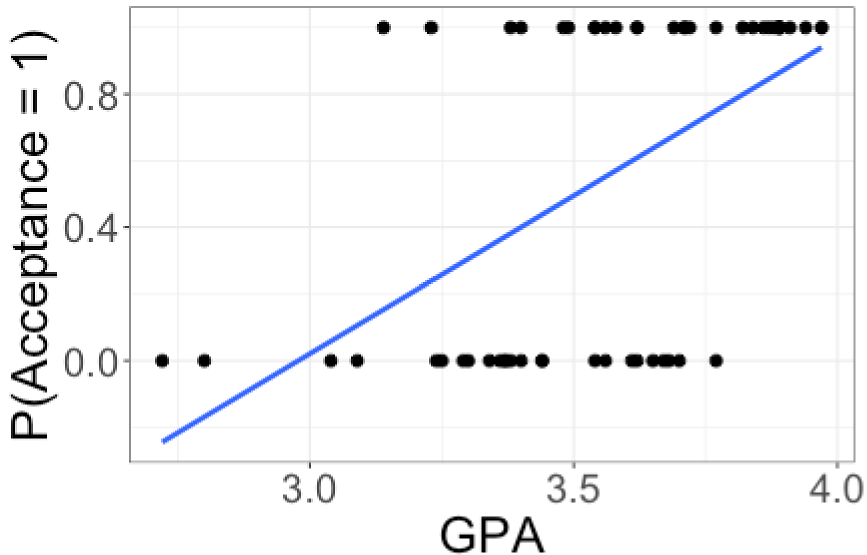
Issues with a binary response



- Acceptance is either 0 or 1 -- we can't have any other values

> How could we change the model to address this issue?

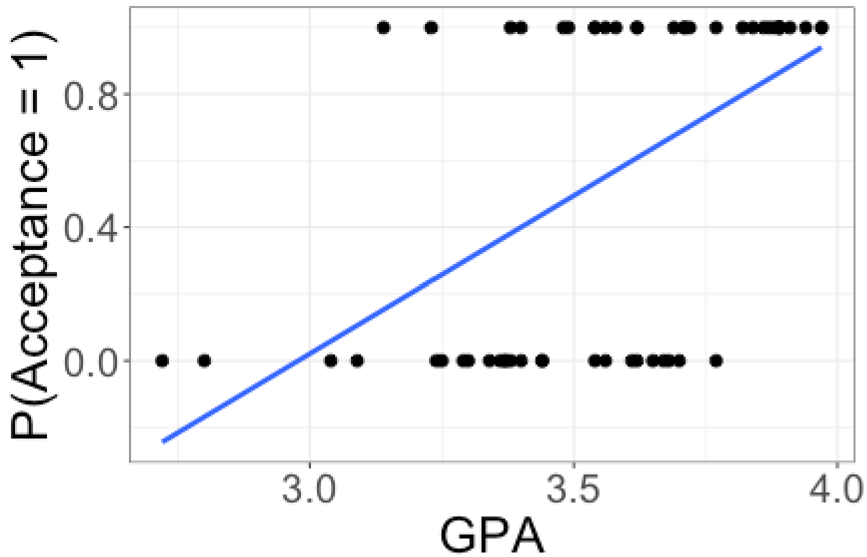
Working with probabilities



- $P(Acceptance = 1)$ = probability student is accepted
- $P(Acceptance = 1) \stackrel{?}{=} \beta_0 + \beta_1 GPA$

> Does this fix our issues with the model?

Issues with probabilities



- probabilities (e.g., $P(Acceptance = 1)$) are between 0 and 1
- Lines are never constrained, unless the slope is 0

Exploring binary outcomes

How can I summarize binary variables?

- Probabilities
- Odds
- Odds ratios

Probabilities

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What is the probability a TMS patient was pain free after two hours?

Probabilities

	TMS	Placebo	Total
Pain free two hours later	39	22	61
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What is the probability a TMS patient was pain free after two hours?

39/100 = 0.39

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
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Total	100	100	200

Comparing probability of being pain free to probability of not being pain free:

- $P(\text{TMS patient is pain free}) = 0.39$
- $P(\text{TMS patient is not pain free}) = 0.61$

Odds that a TMS patient is pain free after two hours:

$$\frac{0.39}{0.61} = 0.64$$

Odds

	TMS	Placebo	Total
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Odds that a TMS patient is pain free after two hours:

$$\frac{0.39}{0.61} = 0.64$$

- The probability of being pain free is 0.64 times the probability of not being pain free, for TMS patients

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Odds that a TMS patient is pain free after two hours:

$$\frac{0.39}{0.61} = 0.64$$

- Odds of being pain free = $\frac{\text{probability pain free}}{1 - \text{probability pain free}}$

Odds

Definition: Let π denote the probability an event happens (e.g., $\pi = P(Acceptance = 1)$). Then the *odds* are

$$\frac{\pi}{1-\pi}$$

- Note that here π is a probability, not the number 3.14159...

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What are the odds a placebo patient was pain free after 2 hours?

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What are the odds a placebo patient was pain free after 2 hours?

$$\frac{0.22}{1 - 0.22} = \frac{0.22}{0.78} = 0.28$$

Odds ratios

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Comparings odds:

- Odds that a TMS patient is pain free after 2 hours: 0.64
- Odds that a placebo patient is pain free after 2 hours: 0.28

Odds ratio: $\frac{0.64}{0.28} = 2.29$

- The odds of being pain free after 2 hours are 2.29 times as high for TMS patients as for placebo patients

Class activity

Work on the class activity (handout).

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

> What was the probability of survival for first class passengers?

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> What was the probability of survival for first class passengers?

$$136/216 = 0.63$$

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> What were the odds of survival for first-class passengers?

$$(136/216)/(80/216) = 136/80 = 1.7$$

> How would I interpret these odds?

Class activity

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> What were the odds of survival for first-class passengers?

$$(136/216)/(80/216) = 136/80 = 1.7$$

> How would I interpret these odds?

For first class passengers, the probability of survival was 1.7 times the probability of not surviving.

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

> Calculate the odds ratio comparing odds of survival for first-class passengers to odds of survival for second and third class passengers.

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

> Calculate the odds ratio comparing odds of survival for first-class passengers to odds of survival for second and third class passengers.

$$1.7/0.44 = 3.86$$

So the odds of survival are 3.86 times higher for first-class passengers than for second and third class passengers

Odds and probabilities

Definition: Let π denote the probability an event happens (e.g., $\pi = P(Acceptance = 1)$). Then the *odds* are

$$\frac{\pi}{1-\pi}$$

- Note that here π is a probability, not the number 3.14159...

> If the probability of an event is > 0.5, what is true about the odds?

(A) Odds > 0.5

(B) Odds > 0

(C) Odds > 1

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Answer: Odds > 1

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> True or false: the odds increase when the probability increases.

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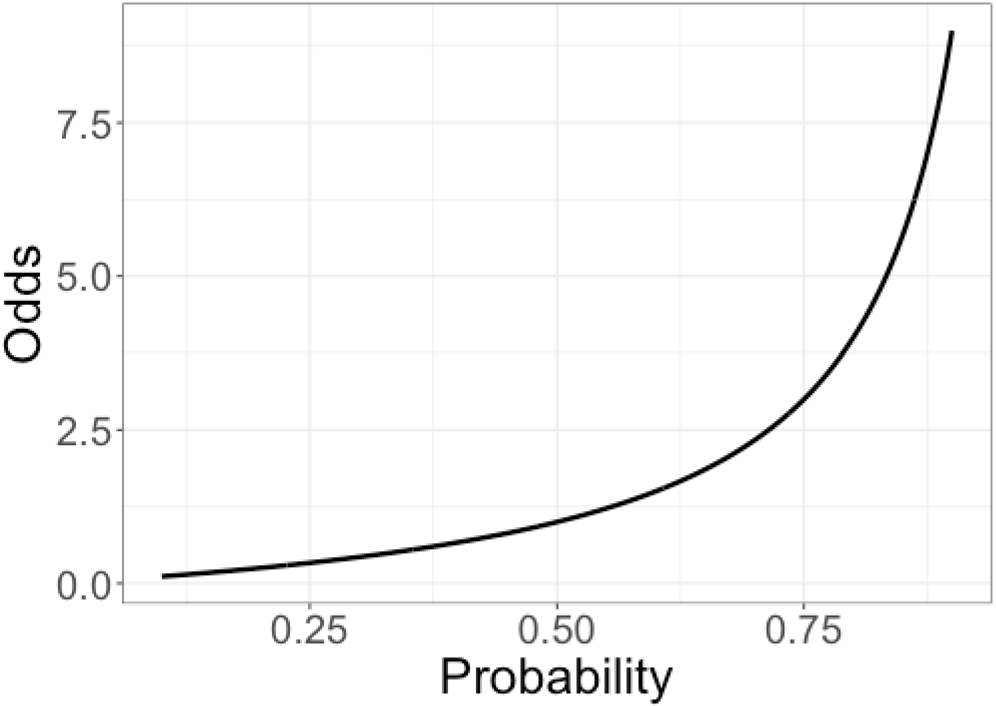
> True or false: the odds increase when the probability increases.

Answer: True

Odds as a function of probability

$$\text{Odds} = \frac{\pi}{1 - \pi}$$



Odds

$$\text{Odds} = \frac{\pi}{1-\pi}$$

Probabilities are between 0 and 1. What range of values can odds take?

(A) between 0 and 1

(B) between 0 and ∞

(C) between $-\infty$ and ∞

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Answer: between 0 and ∞

Log odds

Definition: Let π denote the probability an event happens (e.g., $\pi = P(Acceptance = 1)$). Then the *log odds* are

$$\log\left(\frac{\pi}{1-\pi}\right)$$

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Answer: between $-\infty$ and ∞

Logistic regression

Logistic regression model:

$$\pi = P(Acceptance = 1)$$

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 GPA$$

- log odds are between $-\infty$ and ∞, so we can use a line!

> Where did the ε term go?

Logistic regression

Logistic regression model:

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$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 GPA$$

- log odds are between $-\infty$ and ∞, so we can use a line!

> Where did the ε term go?

- The ε term is used to include randomness in the model
- The probability π *already* incorporates randomness
- The probability and log odds are not random themselves; they describe randomness for the response

Fitted regression model

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \text{GPA}$$

> What are the estimated log odds of acceptance for a student with a 3.0 GPA?

Fitted regression model

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \mathrm{GPA}$$

> What are the estimated log odds of acceptance for a student with a 3.0 GPA?

Answer: $-19.21 + 5.45 \cdot 3 = -2.86$

Fitted regression model

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \text{GPA}$$

> What are the estimated *odds* of acceptance for a student with a 3.0 GPA?

Fitted regression model

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What are the estimated *odds* of acceptance for a student with a 3.0 GPA?

Answer: $\exp\{-2.86\} = 0.0573$